Advanced Relativity Quiz

1. According to special relativity, what happens to time as an object approaches the speed of light?

- Time slows down
- Time speeds up
- Time remains constant
- Time becomes irrelevant

Answer: Time slows down

2. What is the name of the famous equation that relates energy and mass, derived from special relativity?

- E = mc^2
- E = mv^2
- F = ma
- P = mv

Answer: E = mc^2

3. In general relativity, what is the curvature of spacetime caused by?

- The presence of matter and energy
- The absence of matter and energy
- The speed of light
- The laws of motion

Answer: The presence of matter and energy

4. What is the term for the phenomenon where the frequency of light is shifted towards the red end of the spectrum as it escapes from a region with strong gravitational field?

- Gravitational redshift
- Gravitational blueshift
- Cosmological redshift
- Doppler shift

Answer: Gravitational redshift

5. According to the theory of general relativity, what is the predicted phenomenon where the rotation of a massive object 'drags' spacetime around with it?

- Frame-dragging
- Gravitational waves
- Time dilation
- Length contraction

Answer: Frame-dragging